Department of Electrical and Electronics Engineering

Academic Year 2024 – 2025 Odd Semester
Degree, Semester & Branch: III Semester B.E EEE
Course Code & Title: EE3302 – Digital logic circuits
Name of the Faculty member (s): Dr. K. Karthikeyan

Innovative Practice Description

- **Unit / Topic: Unit I / Comparison of RTL, DTL, TTL, ECL and MOS families**
- **Course Outcome:** CO 1
- **Topic Learning Outcome:** TLO 3
- **Activity Chosen: Think-Pair-Share**
- **Justification:**
  Think-Pair-Share is an effective method for this topic because:
  - ✓ It allows students to initially grapple with the complexities of each logic family independently, fostering deeper understanding.
  - ✓ Partner discussions encourage students to articulate their thoughts, clarify misconceptions, and learn from diverse perspectives.
  - ✓ The sharing phase brings valuable insights to the entire class, enriching the overall understanding and promoting a collaborative learning environment.
- **Time Allotted for the Activity:** 15 minutes
- **Details of the Implementation:**
  - ✓ The students will spend 5 minutes individually researching and noting down the key characteristics (speed, power consumption, noise margin, complexity) of each logic family (RTL, DTL, TTL, ECL, MOS).
  - ✓ Then they will discuss for 7 minutes about their findings with a partner, comparing and contrasting the different families. Identifying their strengths, weaknesses, and typical applications.
  - ✓ One student will share the group's key insights with the class for 3 minutes, focusing on the trade-offs involved in choosing one family over another for specific designs.
- **CO – PO / PSO mapping:**

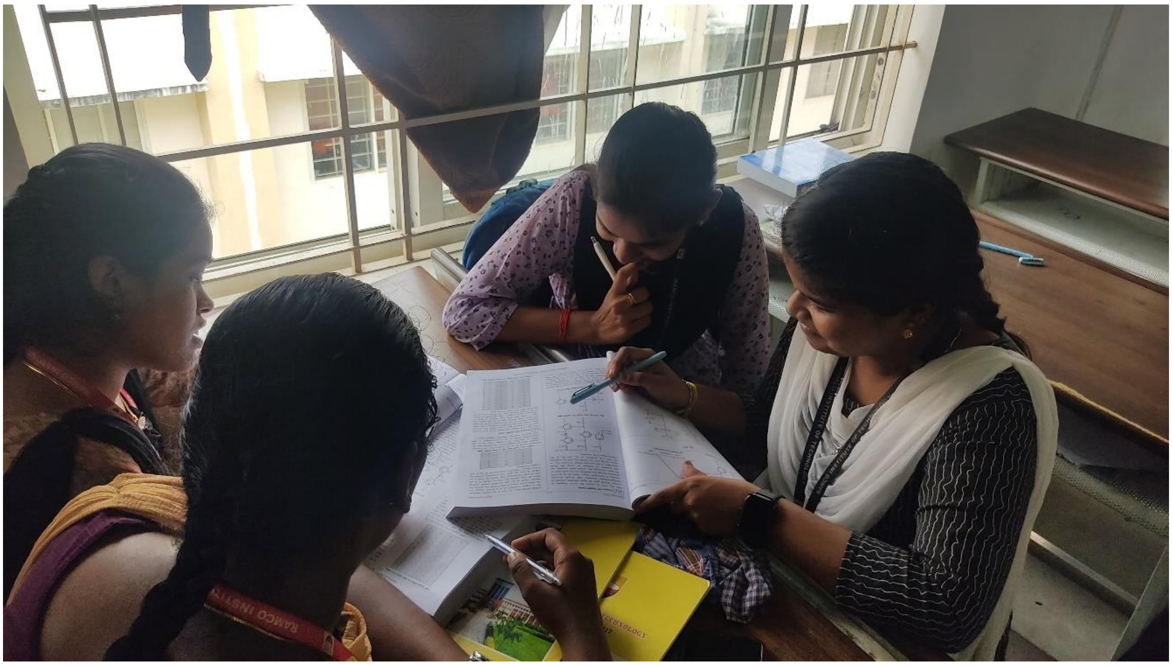
CO	PO1	PO2	PO3	PO4	PO8	PO12	PSO2	PSO3
CO1	3	3	2	1	1	1	2	1

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	**PO9**	**PSO2**
	1	**1**
Justification for correlation	The students are working individually as well as in a group. Hence it is mapped with 1	The students are comparing and discussing the logic families in terms of design perspective for engineering applications. Hence it is mapped with 1

- **Images / Screenshot of the practice:**



- **Reflective Critique:**
  - ❖ ***Feedback of practice from students and other stakeholders:***
    - ✓ It helped the students to understand the material better when they had to explain it to someone else.
    - ✓ The students enjoyed discussing the different logic families with the team. They learned from each other."
    - ✓ The activity made the class more engaging and less passive.
  - ❖ ***Benefit of the practice:*** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)
    - ✓ Students actively engage with the material through individual research and peer discussion.
    - ✓ Deepened understanding of complex concepts through explanation and clarification.
    - ✓ Fostered teamwork and communication skills through partner discussions.
    - ✓ Enhanced ability to articulate and defend their ideas.

- ***Challenges faced in implementation:***
  - ✓ Ensuring sufficient time for each phase (Think, Pair, Share) can be challenging.
  - ✓ Some students may not have been prepared for the discussion.
  - ✓ It can be challenging to ensure equal participation from all students in the pairs.
  - ✓ Some students may feel uncomfortable sharing their ideas in front of the class.

References:

- Morris Mano. M, “Digital Logic and Computer Design”, Prentice Hall of India, 3rd Edition, 2005.
- https://en.wikipedia.org/wiki/Logic_family
- https://www.iitg.ac.in/apvajpeyi/ph218/Lec-28.pdf
- https://lth.engineering.asu.edu/2021/09/think-pair-share/
- https://www.kent.edu/ctl/think-pair-share

Signature of Faculty Member **HoD/EEE**

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Innovative Practice Description

- **Unit / Topic: Unit II / Minimization using K-maps**
- **Course Outcome:** CO 2
- **Topic Learning Outcome:** TLO 5
- **Activity Chosen: Sage and Scribe**
- **Justification:**
  - ✓ This Sage and Scribe approach effectively combines individual learning with collaborative knowledge building, making it ideal for mastering the challenging process of K-map minimization.
  - ✓ This activity is a powerful pedagogical tool for teaching K-map minimization as it promotes active engagement, collaboration, communication, and critical thinking.
  - ✓ It helps students not only grasp the technical aspects of K-map minimization but also develop essential soft skills that are valuable in real-world problem-solving
- **Time Allotted for the Activity:** 15 minutes
- **Details of the Implementation:**
  - ✓ Students are paired. One student is designated the "Sage" (expert) and the other the "Scribe" (note-taker).
  - ✓ Each pair is assigned a K-map minimization problem.
  - ✓ The Sage guides the Scribe through the K-map minimization process, explaining each step clearly and thoroughly.
  - ✓ The Scribe carefully records the steps, diagrams, and explanations provided by the Sage.
  - ✓ After completing one problem, the roles of Sage and Scribe are reversed.
  - ✓ Key insights and common challenges are discussed as a class.
- **CO – PO / PSO mapping:**

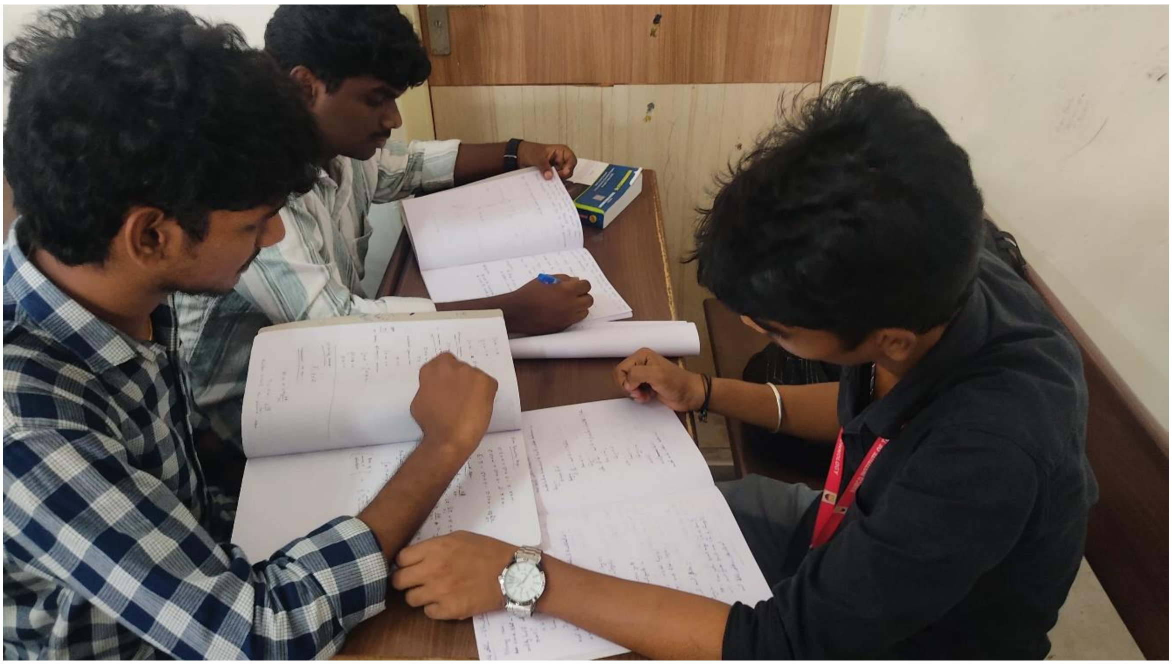
CO	PO1	PO2	PO3	PO4	PO8	PO12	PSO2	PSO3
CO1	3	3	2	1	1	1	2	1

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	**PO9**	**PSO2**
	1	**1**
Justification for correlation	The students are working individually as well as in a group. Hence it is mapped with 1	The students are learning k-map simplification which is used in design process. So, it is mapped with 1.

- **Images / Screenshot of the practice:**



- **Reflective Critique:**
  - ❖ ***Feedback of practice from students and other stakeholders:***
    - ✓ Students generally appreciated the interactive nature of the sage and scribe activity.
    - ✓ The students reported better understanding and retention of K-map minimization concepts through peer explanation and collaboration.
    - ✓ Students appreciated the hands-on approach, finding it engaging and effective in reinforcing concepts. They particularly valued the collaborative discussions and the step-by-step demonstrations by the sage.
    - ✓ Some students noted that switching roles helped them grasp both perspectives.
  - ❖ ***Benefit of the practice:*** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)
    - ✓ The sage and scribe activity involves one student (the sage) explaining the concepts and steps involved in K-map minimization while another student (the scribe) writes down the steps and solution.

- ✓ The scribe provides immediate feedback to the sage, asking questions and seeking clarifications, which reinforces understanding and corrects misconceptions in real-time.
- ✓ Peer discussions fostered teamwork and improved communication skills.
- ✓ The scribe role taught students to record processes systematically, which is crucial for technical fields.

- ❖ ***Challenges faced in implementation:***
  - ✓ Groups often needed more time to complete the activity, especially when concepts were new or complex.
  - ✓ Students initially struggled to understand their roles as sage or scribe, leading to inefficiencies.
  - ✓ Some students felt the pace was too fast during the sage's explanation, making it hard to keep up.
  - ✓ A few also struggled with balancing their roles in group discussions and completing the task.
  - ✓ Variations in students' prior knowledge and confidence levels can affect the effectiveness of the activity. More advanced students may need to be paired carefully with those requiring more support.

References:

- ❖ Morris Mano. M, "Digital Logic and Computer Design", Prentice Hall of India, 3rd Edition, 2005.
- ❖ https://en.wikipedia.org/wiki/Karnaugh_map
- ❖ https://physics.umd.edu/~drew/spr07/Karnaugh%20Maps%20-%20Rules%20of%20Simplification.htm
- ❖ https://krejcicreations.com/4-simple-and-effective-engagement-strategies/

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Innovative Practice Description

- **Unit / Topic: Unit III / Design of synchronous sequential circuits – Moore models**
- **Course Outcome:** CO 3
- **Topic Learning Outcome:** TLO 10
- **Activity Chosen: Pass the Problem**
- **Justification:**
  - ✓ "Pass the problem" encourages active learning and fosters collaborative problem-solving. Students actively engage with the material by designing different parts of the Moore model circuit. This hands-on approach enhances understanding of the design process.
  - ✓ For Moore model design, it allows students to iteratively build and refine state diagrams and transition tables, ensuring peer validation and a deeper understanding of design principles.
  - ✓ This dynamic approach ensures concepts are thoroughly grasped and applied in varied contexts.
- **Time Allotted for the Activity:** 45 minutes
- **Details of the Implementation:**
  - ✓ The students are divided into smaller groups comprising of slow, fast and medium learners.
  - ✓ Each group is provided with one Moore model design problem.
  - ✓ The design task is divided into 4 stages via state diagram design, state table creation, K – map simplification and circuit implementation.
  - ✓ Each group focuses on one stage at a time.
  - ✓ Groups pass their completed stage to the next group.
  - ✓ Finally, one group presents one complete circuit design with reviews, and discusses improvements with the class.
- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO4	PO8	PO12	PSO2	PSO3
CO1	3	3	2	1	1	1	2	1

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	**PO9**	**PSO2**
	1	**1**
Justification for correlation	The students are working individually as well as in a group. Hence it is mapped with 1	The students are learning Moore models which is used in hardware design process. So, it is mapped with 1.

- **Images / Screenshot of the practice:**

- **Reflective Critique:**
  - ❖ ***Feedback of practice from students and other stakeholders:***
    - ✓ **Students** appreciated the interactive nature of the activity, which encouraged collaboration and peer learning.
    - ✓ They found the iterative design and review process enhanced their understanding of Moore models.
    - ✓ The students expressed that It was fun and engaging to work with different people on the same problem.
    - ✓ They learned different approaches to solving the problem from their classmates.
  - ❖ ***Benefit of the practice:*** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)
    - ✓ The activity promoted critical thinking, teamwork, and communication skills.
    - ✓ Encouraged deeper understanding of Moore model principles through iterative learning and feedback.
    - ✓ Exposed students to varied problem-solving approaches, enhancing their design versatility.

- ✓ Promoted active engagement and reduced passive learning, leading to better retention of concepts.

- ❖ ***Challenges faced in implementation:***
  - ✓ Allocating sufficient time for each group to work on the problem and pass it on.
  - ✓ Ensuring all groups complete the activity within the allotted time.
  - ✓ Ensuring all students actively participate in the design process within each group.
  - ✓ Some students struggled with interpreting partially developed diagrams or tables from other groups.

References:

- ❖ Morris Mano. M, "Digital Logic and Computer Design", Prentice Hall of India, 3rd Edition, 2005.
- ❖ https://en.wikipedia.org/wiki/Moore_machine
- ❖ https://mil.ufl.edu/3701/classes/joel/16%20Lecture.pdf
- ❖ https://drexel.edu/teaching-and-learning/resources/active-and-collaborative-learning/collaborative-learning/

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Innovative Practice Description

- **Unit / Topic: Unit IV / CPLD - FPGA**
- **Course Outcome:** CO 4
- **Topic Learning Outcome:** TLO 12
- **Activity Chosen: Mind map**
- **Justification:**
  - ✓ Mind maps simplify the understanding of Complex Programmable Logic Devices (CPLDs) and Field Programmable Gate Arrays (FPGAs) by visually breaking down intricate concepts and their types.
  - ✓ This visual representation enhances memory retention, fosters connections between ideas, and aids in quick recall, making complex information more accessible and engaging.
  - ✓ It fosters active learning, promotes understanding of FPGA types, and encourages collaboration, aiding students in grasping the topic’s complexity and interrelations in a structured way.
- **Time Allotted for the Activity:** 20 minutes
- **Details of the Implementation:**
  - ✓ Using free mind mapping software (e.g., XMind, MindMeister), students create a mind map starting with "CPLD and FPGA" as the central node.
  - ✓ Branch out to "Definition," "Features," "Applications," and "Types." Under "Types," include FPGA variations like SRAM-based and flash-based.
  - ✓ Highlight differences, unique features, and real-world applications for comprehensive, visually engaging learning.
  - ✓ Students collaboratively refine the map to ensure comprehensive coverage.
  - ✓ Include brief descriptions or key features for each concept.
  - ✓ Use colors, shapes, and images to enhance the visual appeal.

- **CO – PO / PSO mapping:**

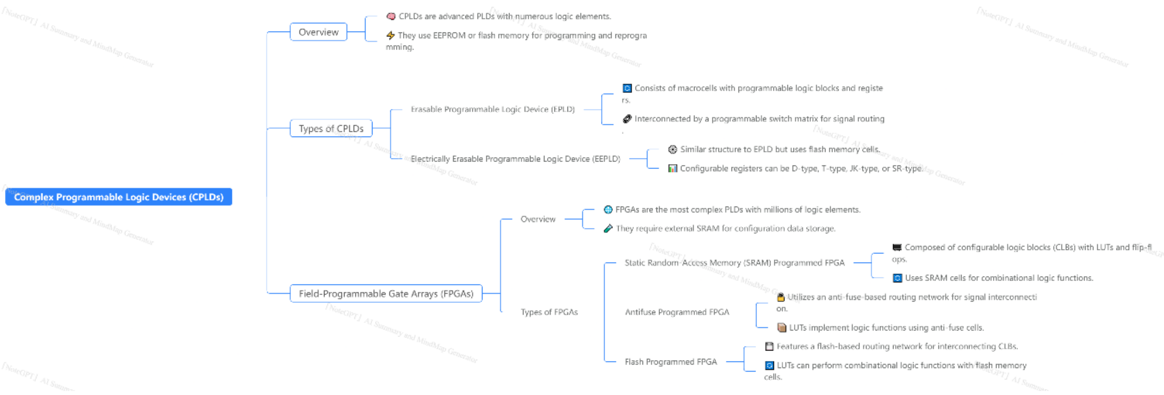
CO	PO1	PO2	PO3	PO4	PO8	PO12	PSO2	PSO3
CO1	3	3	2	1	1	1	2	1

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	PO5	PSO2
	1	1
Justification for correlation	The students are using free mind mapping software for this activity. Hence it is mapped with 1.	The students are learning about CPLD and FPGA which is mostly in digital system hardware design. Hence it is mapped with 1.

- **Images / Screenshot of the practice:**



- **Reflective Critique:**
  - ***Feedback of practice from students and other stakeholders:***
    - ✓ Students found the mind map activity engaging and visually appealing, aiding in understanding the complex concepts of CPLDs and FPGAs.
    - ✓ They also appreciated the method for its clarity and ability to foster connections between ideas, making the learning experience more interactive and memorable.
    - ✓ They expressed that the activity made their learning more engaging and less monotonous.
    - ✓ They suggested additional guidance for first-time users of the software.
  - ***Benefit of the practice:*** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)
    - ✓ The activity improved students' conceptual understanding and retention by transforming complex topics into a structured, visual representation.
    - ✓ It fostered collaboration, creativity, and active learning, making the technical subject more accessible.
    - ✓ Additionally, the visual format catered to various learning styles, making complex topics accessible.
  - ***Challenges faced in implementation:***
    - ✓ Challenges included initial unfamiliarity with mind mapping software, requiring some technical guidance.
    - ✓ Additionally, some students needed time to adapt to this visual learning approach, and ensuring the accuracy and completeness of the mind maps required careful oversight.
    - ✓ Some students struggled with organizing extensive information effectively within the tool, requiring additional instruction and time to optimize their contributions.
    - ✓ Despite these hurdles, the activity proved beneficial overall.

References:

- Morris Mano. M, “Digital Logic and Computer Design”, Prentice Hall of India, 3rd Edition, 2005.
- https://www.electrical4u.com/programmable-logic-devices/
- https://en.wikipedia.org/wiki/Mind_map
- https://notegpt.io/

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Innovative Practice Description

- **Unit / Topic: Unit V / VHDL Operators**
- **Course Outcome:** CO 5
- **Topic Learning Outcome:** TLO 14
- **Activity Chosen: One-minute paper**
- **Justification:**
  - ✓ The one-minute paper encourages students to reflect on their understanding of VHDL operators quickly, promoting active engagement.
  - ✓ This activity prompts immediate reflection, reinforcing key concepts and identifying knowledge gaps.
  - ✓ It helps identify key learning points and areas of confusion, providing immediate feedback for both students and instructors.
  - ✓ This simple, time-efficient activity fosters critical thinking and reinforces core concepts effectively.
- **Time Allotted for the Activity:** 15 minutes
- **Details of the Implementation:**
  - ✓ At the end of the session on VHDL operators, I have asked the students to write down key takeaways, unresolved doubts, and practical applications.
  - ✓ They wrote concise responses within one minute.
  - ✓ I have collected the papers from the students. Quickly reviewed the papers to identify common questions, misconceptions, and areas where further clarification is needed.
  - ✓ I have addressed common questions and misconceptions in the next class.
- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO4	PO5	PO8	PO12	PSO1	PSO2	PSO3
CO1	3	3	2	1	1	1	1	2	2	1

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	**PO9**	**PSO3**
	1	**1**
Justification for correlation	The students are working individually as well as in a group. Hence it is mapped with 1	The students are learning new software package for writing their code. So, it is mapped with 1.

- **Images / Screenshot of the practice:**

- **Reflective Critique:**
  - ❖ ***Feedback of practice from students and other stakeholders:***
    - ✓ Students appreciated the opportunity to summarize key learnings and express doubts quickly.
    - ✓ They valued the activity's simplicity and effectiveness in gauging comprehension but suggested integrating digital tools for efficiency.
    - ✓ They also expressed that it provides immediate insight into their understanding and misconceptions, allowing for quick adjustments to subsequent lessons.
  - ❖ ***Benefit of the practice:*** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)
    - ✓ The activity provided immediate insights into students' understanding of VHDL operators, enabling targeted feedback.
    - ✓ It adopted active reflection, reinforced learning, and allowed me to identify common misconceptions efficiently.
    - ✓ This active engagement helps in identifying gaps in knowledge and promotes active participation and engagement in the learning process.

- ***Challenges faced in implementation:***
    - ✓ Time constraints limited detailed responses, and some students struggled to articulate their thoughts within the given time.
    - ✓ Ensuring meaningful participation in larger classes required additional effort to review all submissions promptly.

References:

- Morris Mano. M, “Digital Logic and Computer Design”, Prentice Hall of India, 3rd Edition, 2005.
- https://peterfab.com/ref/vhdl/vhdl_renerta/mobile/source/vhd00047.htm
- https://www.rochester.edu/college/teaching/teaching-guidance/one-minute-paper.html.
- https://www.westernsydney.edu.au/__data/assets/pdf_file/0020/1133363/one_minute_paper_1.pdf.

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